Throughout, let $(M^n, g, \alpha)$ be an oriented Riemannian manifold with $\alpha$ a **closed** strongly nondegenerate $k$-form, where strongly nondegenerate means $(X_1 \wedge \ldots \wedge X_{k-1}) \lrcorner \alpha = 0$ implies $X_1 \wedge \ldots \wedge X_{k-1} = 0$ for any $X_1, \ldots, X_{k-1} \in \mathfrak{X}(M)$.

Introduction to $G_2$-geometry! here

Instantons: preserved by vector cross product
Branes: maximal dimension on which $\alpha$ vanishes.

| Manifolds $M$ ($\dim M$) | VCP form $\phi$ (degree of $\phi$) | Instanton $A$ ($\dim A$) | Brane $C$ ($\dim C$) |
|---|---|---|---|
| Oriented Riem. mfd. ($n$) | Volume form ($n$) | Open subset ($n$) | Hypersurface ($n-1$) |
| Kähler manifold ($2m$) | Kähler form (2) | Holomorphic curve (2) | Lagrangian submanifold ($m$) |
| $G_2$-manifold (7) | $G_2$-form (3) | Associative submanifold (3) | Coassociative submanifold (4) |
| $Spin(7)$-manifold (8) | Cayley form (4) | Cayley submanifold (4) | N/A |

**Proposition** (Brown-Gray '67)
Up to rescaling, the only strongly nondegenerate forms are almost symplectic forms, volume forms, (possibly split) associative 3-forms, and (possibly split) Cayley 4-forms.
*Sketch:*
It suffices to prove this linear algebraically on $\mathbb{R}^n$. By lowering an index, a strongly nondegenerate form is the same thing as a vector cross product

$$\chi : \Lambda^{k-1}TM \to TM$$

which is a form that satisfies

$$g(\chi(X_1, \ldots, X_{k-1}), X_i) = 0$$

for all $1 \leq i \leq k-1$ and $|\chi(X_1, \ldots, X_{k-1})|^2 = |X_1 \wedge \ldots \wedge X_{k-1}|^2$.

Letting $V_{k-1,n}$ denote the Stiefel manifold of orthonormal $(k-1)$-tuples in $\mathbb{R}^n$, this induces a section

$$(X_1, \ldots, X_{k-1}) \mapsto \left(X_1, \ldots, X_{k-1}, \frac{\chi(X_1, \ldots, X_{k-1})}{|\chi(X_1, \ldots, X_{k-1})|}\right)$$

of the projection $V_{k,n} \to V_{k-1,n}$. By algebraic topology (Eckmann, 1963, Stetige Lösungen linearer Gleichungssysteme), this can only occur if $k \in \{2, n\}$, or $(k, n) = (3, 7), (4, 8)$. We are then done unless we are in the exceptional $(3, 7)$ or $(4, 8)$ cases, where one must then show that the bilinear forms induced by the SND form are nondegenerate, giving the remaining cases. ■

By the previous proposition, assuming we are not in the split case (so induced bilinear form from SND is a metric), we may assume $\alpha, g$ are compatible in the sense that $\chi$ of the previous proof is a vector cross product (i.e. $|\chi(X_1, \ldots, X_{k-1})|^2 = |X_1 \wedge \ldots \wedge X_{k-1}|^2$ in the notation of the above proof), and that $\alpha$ is a calibration wrt $g$.
To show this makes $\alpha$ into a calibration, let $e_1, \ldots, e_k$ be orthonormal. Then, because $\chi$ is a vector cross product,

$$|\alpha(e_1, \ldots, e_k)| = |g(\chi(e_1 \wedge \ldots \wedge e_{k-1}), e_k)| \leq |\chi(e_1 \wedge \ldots \wedge e_{k-1})| \, |e_k| = 1$$

with equality achieved by taking $e_k = \chi(e_1 \wedge \ldots \wedge e_{k-1})$ courtesy of our scaling.
**Definition**
The (free) loop space $LM$ is the space of smooth maps $S^1 \to M$.[1]

[1] If one speaks of the pointed loop space, they mean the loop space where the maps are basepointed (often, algebraic topologists mean the pointed loop space when they say loop space).

Its subset (which will be open in its manifold topology) of singular knots, denoted $\widehat{\mathcal{K}}$, consists of the immersions $\gamma : S^1 \to M$ that, for some finite set $A$ (depending on $\gamma$), induce an embedding of $S^1 \setminus A$ such that for all $x_1 \neq x_2 \in A$, $\gamma(x_1), \gamma(x_2)$ are tangent to only finite order[2].
The (open) subset $\mathcal{K}$ of $\widehat{\mathcal{K}}$ of embeddings is called the knot space of $M$.

**Remark**
Brylinski only considers the case $n = 3$, $\alpha = \mathrm{vol}$, but many of his results hold when $n \neq 3$. In turn, he gets a Kähler package as we will see below. However, his arguments concerning the group of diffeomorphisms preserving $\alpha$ (in his case, volume-preserving diffeomorphisms) can, with some work, be modified to work in our case as I have done at the bottom of this document, though certain parts of his argument required explicit calculations in terms of the 3-dimensional cross product and curl, which I have replaced with more general computations that work for both our case and his.
Lee-Leung consider any $(M^n, g, \chi, \alpha)$ as above, but instead take maps $\Sigma \to M$ where $\Sigma$ is a compact manifold of the appropriate dimension to make $\beta$ as defined below a 2-form. Some of Lee-Leung's arguments still apply (with routine modifications) in the case below where we always take $S^1$, namely the "localization" arguments when certain forms vanish (how we show $\beta$'s closure implies $\alpha$'s closure, and the Proposition below concerning branes). Their arguments concerning instanton submanifolds do not seem to generalize when $\beta$ is not a symplectic form.

The topology and smooth structure on $LM$ (and by openness also on $\widehat{\mathcal{K}}$ and $\mathcal{K}$) are given as follows: $LM$ is locally modelled by the Frechét space $\mathcal{C}^\infty(S^1, \mathbb{R}^n)$. Its topology is given by the inverse limit of Hilbert spaces $H^k(S^1, \mathbb{R}^n)$ (taking the limit over $k$), which is the completion of $\mathcal{C}^\infty(S^1, \mathbb{R}^n)$ with respect to the seminorm

$$\|f\|_k = \int_0^1 \sum_{j=0}^{k} |f^{(j)}(t)|^2 dt$$

where throughout we choose a periodic coordinate $t$ on the circle so that $S^1$ has length 1.
We can get a chart containing a loop $\gamma : S^1 \to M$ as follows. Since any strongly nondegenerate form induces an orientation, by pulling back via $\gamma$ we get an oriented bundle $\gamma^*TM \to S^1$. Now, since $\gamma(S^1)$ is compact, there is $\varepsilon > 0$ small enough so that the exponential map $\exp_{\gamma(t)}$ of $g$ is a diffeomorphism from $B_\varepsilon(0) \subseteq T_{\gamma(t)}M$ onto some open subset $U_t$ of $\gamma(t)$ in $M$ for all $t \in S^1$, so that the image of $\exp_{\gamma(t)}$ consists of those loops $\Xi$ with $\Xi(t) \in U_t$ for all $t$, which is a neighbourhood of $\gamma$ (those loops whose images lie in a small enough tubular neighbourhood of $\gamma(S^1)$.)

Let $\gamma^*TM_\varepsilon$ be tangent vectors of length less than $\varepsilon$, which is open in $\gamma^*TM$. Then, the exponential map of $g$ gives a map $E : \Gamma(S^1, \gamma^*TM_\varepsilon) \to LM$, with $E(X)_t = \exp_{\gamma(t)}(X_t)$, and this is a bijection onto its image because of how we chose $\varepsilon$. However, since $\gamma^*TM$ is an oriented bundle over $S^1$, it is trivial and thus we have a trivialization $\tau : \gamma^*TM \xrightarrow{\cong} S^1 \times \mathbb{R}^n$. Composing, this gives a chart

$$\alpha \mapsto \mathrm{pr}_2 \circ \tau \circ E^{-1}(\alpha) \in \mathcal{C}^\infty(S^1, \mathbb{R}^n)$$

for any $\alpha$ in the image of $E$. Its inverse sends $u \in \mathcal{C}^\infty(S^1, \mathbb{R}^n)$ to the loop $t \mapsto \exp_{\gamma(t)}(\tau^{-1}(t, u(t)))$.

**Proposition** (Brylinski 3.1.2, Pressley-Segal chapter 3)
The above charts give a smooth paracompact Frechét manifold structure for $LM$. Moreover, the topology and smooth structure are independent of the metric $g$.

By the above description, $T_\gamma\mathcal{K} = \Gamma(S^1, \gamma^*TM)$.

[2]Which means that for $x_1 \neq x_2$ such that $\gamma(x_1) = \gamma(x_2)$ there is some finite $\ell \in \mathbb{N}$ such that the $\ell$-jets of the two one-sided component curves of $\gamma$ approaching $\gamma(x_1) = \gamma(x_2)$ are not equal. That is, the one-sided Taylor expansions of $\gamma$ approaching $\gamma(x_1) = \gamma(x_2)$ can agree only to finite order.

Now, the orientation-preserving diffeomorphisms $\mathrm{Diff}^+(S^1)$ act on the right on $LM$ by composition (and preserves $\widehat{\mathcal{K}}, \mathcal{K}$).

**Proposition** (Brylinski 3.1.3 and 3.1.4)
The quotient $\mathcal{M} = \mathcal{K}/\mathrm{Diff}^+(S^1)$ is a (infinite-dimensional) smooth manifold, locally modelled by $\mathcal{C}^\infty(S^1, \mathbb{R}^{n-1})$. In particular, its tangent spaces are given by $T_{[\gamma]}\mathcal{M} = \Gamma(S^1, N_{S^1/M})$, where $N_{S^1/M}$ is the normal bundle to any choice of immersion $\gamma \in \mathcal{K}$.
*Sketch:*
Intuitively, given the above tubular neighbourhoods that we use to make charts, quotienting out $\mathrm{Diff}^+(S^1)$ kills the tangent direction and leaves the normal directions for charts on $\mathcal{M}$, and things are smooth since $\mathrm{Diff}^+(S^1)$ acts freely on $\mathcal{K}$ because $\gamma \circ \phi = \gamma$ for $\phi \in \mathrm{Diff}^+(S^1)$ forces $\phi = \mathrm{id}$ by injectivity of $\gamma$.

More precisely, given $\gamma \in \mathcal{K}$, let $N$ be the normal bundle to $\gamma$ and let $P \subseteq N$ be a tubular neighbourhood of $\gamma(S^1)$ such that there is an embedding $i : P \to M$ extending $\gamma$, and let $p : P \to S^1$ be the projection. Let $U$ be the open subset of $LP = \mathcal{C}^\infty(S^1, P)$ consisting of maps $\delta : S^1 \to P$ such that $p \circ \delta \in \mathrm{Diff}^+(S^1)$. Then $\delta \mapsto i \circ \delta$ is a smooth map $U \to \mathcal{K}$. This is a local diffeomorphism, hence an immersion, since $i^{-1}$ is smooth on $i(P)$, so for those $\epsilon \in \mathcal{K}$ with $\epsilon(S^1) \subseteq i(P)$, we have an inverse map $\epsilon \mapsto i^{-1} \circ \epsilon$. Moreover, its image is open in $\mathcal{K}$. To see this, note that its image is the set of maps $\epsilon : S^1 \to M$ such that $\epsilon(S^1) \subseteq i(P)$ and $p \circ (i^{-1} \circ \epsilon) \in \mathrm{Diff}^+(S^1)$, and both conditions are open conditions: the former because compactness of $\epsilon(S^1)$ tells us that its distance to $M \setminus i(P)$ is positive, and thus $\mathcal{C}^\infty$-small perturbations keep things inside $i(P)$, and the latter as seen previously. This allows us to identify $U$ with an open subset of $\mathcal{K}$, and by definition of $U$, it is stabilized by the $\mathrm{Diff}^+(S^1)$-action. Now, let $V$ be the closed subset of $U$ consisting of those $\delta : S^1 \to P$ such that $p \circ \gamma = \mathrm{id}$, that is, the space of sections of $p$, which is a closed submanifold of $U$ as it is the preimage of $\{\mathrm{id}\}$ under the map $U \to LS^1, \delta \mapsto p \circ \delta$. Then, we have a smooth action $V \times \mathrm{Diff}^+(S^1) \to U$ given by composition on the left, and it is a diffeomorphism because we have a (smooth) inverse map $\delta \mapsto (\delta \circ (p \circ \delta)^{-1}, p \circ \delta)$. All-in-all, this gives a diffeomorphism $U/\mathrm{Diff}^+(S^1) \xrightarrow{\cong} V$, which is our chart on $\mathcal{K}$ containing $[\gamma]$. ■

In what follows, we will be doing the usual calculus on manifolds over the infinite dimensional manifolds $\mathcal{M}$ and $\mathcal{K}$. Because $\mathcal{K}$ and $\mathcal{M}$ are built from spaces of smooth maps, which are quite nice, most things work as they do in finite dimensions. In particular, per Pressley-Segal section 3.1, differential forms and the exterior derivative behave as usual, and the Poincaré Lemma and de Rham's theorem still apply. Smooth vector fields are still defined as sections of the tangent bundle, and act as differentiation operators in the same way, and in turn their Lie bracket has the same definition. However, flows of vector fields need not exist in general.

We define on $\mathcal{K}$ a form $\beta$ by integrating our original form $\alpha$ along the fibers of the evaluation map $\mathrm{ev} : \mathcal{K} \times S^1 \to M$. Since $S^1$ is 1-dimensional and $\alpha$ is a $k$-form, this yields a $(k-1)$-form on $\mathcal{K}$. Explicitly, we have

$$\beta = \int_{S^1} \mathrm{ev}^*\alpha := (\pi_1)_*\mathrm{ev}^*\alpha$$

where $\pi_1 : \mathcal{K} \times S^1 \to \mathcal{K}$ is projection. Consequently, if $X_1, \ldots, X_{k-1} \in T_\gamma\mathcal{K} = \Gamma(S^1, \gamma^*TM)$, then

$$\beta(X_1, \ldots, X_{k-1}) = \int_{S^1} (X_1 \wedge \ldots \wedge X_{k-1}) \lrcorner\, \alpha = \int_0^1 \alpha_{\gamma(t)}(\frac{d}{dt}\gamma(t), X_1(t), \ldots, X_{k-1}(t))dt \quad (\dagger)$$

Since $\alpha$ is closed, we can "differentiate under the integral sign" (Proposition 6.14.1 of Bott-Tu, whose proof works in this context with minor modifications as it goes by differentiating ($\dagger$) under the integral sign and patching everything together with partitions of unity, which are both still valid in this setting) to get

$$d\beta = \int_{S^1} d(\mathrm{ev}^*\alpha) = \int_{S^1} \mathrm{ev}^*(d\alpha) = 0$$

The converse is also true, via an argument called "localization":

**Proposition**
If $\beta$ is closed, then $\alpha$ is closed.
*Proof:*
We need to show that for the $(k+1)$-form $d\alpha$, we have $(d\alpha)_p(X_1,\ldots,X_{k+1})=0$ for any tangent vectors $X_1,\ldots,X_{k+1}$ at $p$. Without loss of generality, suppose $X_1,\ldots,X_{k+1}$ are orthonormal. Choose $\gamma\in\mathcal{K}$ with $p\in\gamma(S^1)$ such that $X_1$ is tangential, and $X_2,\ldots,X_{k+1}$ are normal, and extend them to sections in $\Gamma(S^1,\gamma^*TM)$. Now, multiply $X_2$ with a sequence of mollifiers approaching the Dirac delta at $p$, so that we get sections $\widetilde{X}_\varepsilon$ which converge to $\delta(p)(X_2)_p$ as $\varepsilon\to 0$, giving

$$(d\alpha)_p(X_1,\ldots,X_{k+1})=\lim_{\varepsilon\to 0}\left(\int_{S^1}\mathrm{ev}^*(d\alpha)\right)(\widetilde{X}_\varepsilon,X_3,\ldots,X_{k+1})$$

Since $\int_{S^1}\mathrm{ev}^*(d\alpha)=0$ by the assumption on $\beta$, we obtain $d\alpha=0$. ■

The form $\beta$ on $\mathcal{K}$ is basic with respect to the action of $\mathrm{Diff}^+(S^1)$, which means $X\lrcorner\,\beta=0$ and $X\lrcorner\, d\beta=0$ for any vertical vector field $X$ (vertical with respect to $\pi:\mathcal{K}\to\mathcal{M}$). Since $d\beta=0$, this follows from (†) as $\frac{d\gamma}{dt}$ and $f\frac{d\gamma}{dt}$ are parallel for any smooth function $f$.

Alternatively, one can use that being basic is equivalent to being invariant under $\mathrm{Diff}^+(S^1)$ and $X\lrcorner\,\beta=0$ for all vertical vector fields $X$ (i.e. $\beta$ is horizontal). To prove this equivalence, notice that our definition of basic above is equivalent to saying $\beta$ is horizontal and $\mathcal{L}_X\beta=0$ for every vertical $X$ by Cartan's formula. Since $\mathrm{Diff}^+(S^1)$ is connected (so that the fibers of $\pi$ are connected, as they are all diffeomorphic to $\mathrm{Diff}^+(S^1)$), this integrates to say that $\beta$ is invariant under the action of $\mathrm{Diff}^+(S^1)$, and conversely being horizontal and invariant tells us $X\lrcorner\,\beta=0, X\lrcorner\, d\beta=0$.

To check invariance explicitly, let $R_\psi$ be the action by $\psi$ on $\mathcal{K}$ for $\psi\in\mathrm{Diff}^+(S^1)$. Since $(dR_\psi(X))(t)=X_{\psi(t)}$ and $(\gamma\circ\psi)'(t)=\psi'(t)\gamma'(\psi(t))$, we have

$$(R_\psi^*\beta)_\gamma(X_1,\ldots,X_{k-1})=\int_0^1\alpha((\gamma\circ\psi)'(t),X_1(\psi(t)),\ldots,X_{k-1}(\psi(t)))dt$$
$$=\int_0^1\alpha(\gamma'(\psi(t)),X_1(\psi(t)),\ldots,X_{k-1}(\psi(t)))\psi'(t)dt$$

and changing variable $s=\psi(t)$ (using the fact that $\psi$ is oriented) we get the desired invariance. Therefore, $\beta$ descends to a well-defined smooth form on $\mathcal{M}$, which I still denote by $\beta$.

Moreover, we can define the usual $L^2$-metric on $\mathcal{K}$

$$g_\gamma(u,v)=\int_{\gamma(S^1)}g(u,v)\mu_{\gamma(S^1)}=\int_0^1 g(u(t),v(t))|\gamma'(t)|dt$$

which we see descends to a metric on $\mathcal{M}$, as the above integral is independent of reparameterization.

Now, define a vector cross product $C$ (which will take $k-2$ inputs) as follows. For any $[\gamma]\in\mathcal{M}$ and $X_1,\ldots,X_{k-2}\in\Gamma(S^1,N_{S^1/M})$, let $E$ be a unit vector field (for generalization purposes, need to take an oriented orthonormal basis) in $TS^1$ (under the given embedding) and let

$$C(X_1,\ldots,X_{k-2})=\chi(E,X_1,\ldots,X_{k-2})$$

Since the tangent vectors to $\mathcal{M}$ are normal to $S^1$, we are really just taking the previous cross product and contracting out the tangent direction to go down a degree, which since $E$ is unit, gives a vector cross product.

**Proposition**
This vector cross product $C$ is the associated cross product of $\beta$. Accordingly, $\beta$ is strongly nondegenerate.
*Proof:*
Let the notation be as above. We have

$$\alpha(E, X_1, \ldots, X_{k-1})\mu_{S^1} = g(\chi(E, X_1, \ldots, X_{k-2}), X_{k-1})\mu_{S^1} = g(C(X_1, \ldots, X_{k-2}), X_{k-1})\mu_{S^1}$$

so that integrating over $S^1$, by definition of $(X_1 \wedge \ldots \wedge X_{k-1}) \lrcorner \alpha$ we obtain $\beta(X_1, \ldots, X_{k-1}) = g(C(X_1, \ldots, X_{k-2}), X_{k-1})$. (This argument actually shows $C$ is a vector cross product on $\mathcal{M}$ if and only if $\chi$ is a vector cross product on $M$). ■

**Proposition** (pretty sure this is true)
A submanifold $Y$ of $M$ is a brane with respect to $\alpha$ (that is, a maximal-dimension submanifold where $\alpha$ vanishes, which is necessarily of dimension $(n+k-2)/2$) if and only if the submanifold $\mathcal{Y}$ of $\mathcal{M}$ given by the same construction as above restricted to $\mathcal{Y}$ is a maximal subset on which $\beta$ vanishes.
*Proof:*
In one direction, assume $\mathcal{Y}$ is a maximal submanifold on which $\beta$ vanishes. Using that $\beta$ vanishes on $X_1, \ldots, X_{k-1} \in \Gamma(S^1, N_{S^1/Y})$ and applying the localization argument of a previous proposition, we obtain $\alpha(X_1, \ldots, X_{k-1}, E) = 0$ for any unit vector field $E$ of $TS^1$. Varying $X_1, \ldots, X_{k-1}$ and $\gamma$ to cover $TY$, we obtain that $\alpha$ vanishes on $Y$. Moreover, $Y$ is maximal-dimension since $\mathcal{Y}$ is maximal; if there were some submanifold $Y' \subseteq M$ of bigger dimension, then $\beta$ vanishes on $\mathcal{Y}'$ (defined analogously to $\mathcal{M}$ and $\mathcal{Y}'$) and contains (equivalence classes of) knots whose images do not lie in $Y$, because $Y'$ is of bigger dimension (so $Y'$ must necessarily have some points not contained in $Y$, and taking a small enough knot based at one of those points will not lie in $Y$). Then $\mathcal{Y} \cup \mathcal{Y}'$ is a set on which $\beta$ vanishes identically, and strictly contains $\mathcal{Y}$, contradicting maximality.
Conversely, if $Y$ is a brane, then $\beta$ vanishes on $\mathcal{Y}$. Now, suppose $\mathcal{Y}$ is not maximal, so there is some $\mathcal{Z} \supsetneq \mathcal{Y}$ on which $\beta$ vanishes. Then there is a section $V$ in $\Gamma(S^1, N_{S^1/M})$ but not $\Gamma(S^1, N_{S^1/Y})$ with which $\alpha$ vanishes (via the localization argument TODO WRITE OUT) on the linear space spanned by $V_x$ and $T_x\gamma(S^1)$ for any $x \in \gamma(S^1) \subseteq Y$, contradicting maximality of $Y$.

**Corollary**
If $(M, \alpha)$ is a Spin(7)-manifold, then $\mathcal{M}$ has no maximal submanifolds on which $\beta$ vanishes (which are the analogue of coassociative submanifolds).
*Proof*:
Spin(7)-manifolds have no branes. ■

Now, let $G$ be the subgroup of diffeomorphisms of $M$ that preserve $\alpha$, and let $\mathfrak{g}$ be its Lie algebra. Then the group $G$ acts on $\mathcal{K}$ on the left (smoothly), and commutes with the action of $\text{Diff}^+(S^1)$, and so descends to $\mathcal{M}$.

**Proposition**
The action of $G$ preserves $\beta$.
*Proof:*
A simple calculation, which we can do on $\mathcal{K}$ since the actions commute. Let $\phi \in G$ be arbitrary. Then,

$$(\phi^*\beta)_\gamma(X_1, \ldots, X_{k-1}) = \int_0^1 \alpha((\phi \circ \gamma)'(t), d\phi(X_1(t)), \ldots, d\phi(X_{k-1}(t)))dt$$

$$= \int_0^1 \alpha(d\phi(\gamma'(t)), d\phi(X_1(t)), \ldots, d\phi(X_{k-1}(t)))dt = \int_0^1 (\phi^*\alpha)(\gamma'(t), X_1(t), \ldots, X_{k-1}(t))dt$$

$$\overbrace{=}^{\phi^*\alpha=\alpha} \int_0^1 \alpha(\gamma'(t), X_1(t), \ldots, X_{k-1}(t))dt = \beta_\gamma(X_1, \ldots, X_{k-1})$$

as needed. ■

**Theorem**
Assume $k = 3$, so that $(M^n, \alpha)$ is either an oriented 3-manifold or a $G_2$-manifold, and assume $b^2(M) = 0$. Then, for any $X \in \mathfrak{g}$, the map $\langle \mu(\gamma), X \rangle = \int_{\gamma(S^1)} \eta$ is a moment map for the action of $G$, where $\eta$ is a 1-form such that $d\eta = X \lrcorner \alpha$.
*Proof:*
Throughout, let $\mu_X$ denote the map $\langle \mu(-), X \rangle$, and we use $\times$ for the cross product. Note that $X \lrcorner \alpha$ is closed, since $X \in \mathfrak{g}$ implies $\mathcal{L}_\xi \alpha = 0$, so that $d(X \lrcorner \alpha) = -X \lrcorner (d\alpha) = 0$ by Cartan's magic formula. Then, using the fact we may exchange integration and exterior differentiation, we have

$$\langle X \lrcorner \beta, V \rangle = \beta(X, V) = \int_0^1 \alpha(\gamma'(t), X, V)dt = -\int_0^1 X \lrcorner \alpha(\gamma', V)dt = \int_{\gamma(S^1)} V \lrcorner (X \lrcorner \alpha) = \langle d\mu_X, V \rangle$$

so that we have a (possibly non-equivariant) moment map. To check equivariance, we must show $\{X, Y\} = \mu_{[X,Y]}$, where $\{\cdot, \cdot\}$ is the Poisson bracket. Let $F = \mu_X$, $G = \mu_Y$. Then,

$$\{F, G\} = \beta(Y, X) = \int_0^1 \alpha(\gamma'(t), Y, X)dt = \int_0^1 \langle Y \times X, \gamma'(t) \rangle dt$$

On the other hand,

$$\begin{aligned} d((X \times Y)^\flat) &= d(Y \lrcorner X \lrcorner \alpha) = \mathcal{L}_Y(X \lrcorner \alpha) - Y \lrcorner d(X \lrcorner \alpha) \\ &= [Y, X] \lrcorner \alpha + \cancel{X \lrcorner \mathcal{L}_Y \alpha} - Y \lrcorner (\cancel{\mathcal{L}_X \alpha} - \cancel{X \lrcorner d\alpha}) \end{aligned}$$

By definition of $\mu$ we therefore get that $\mu_{[X,Y]}$ integrates $Y \times X$ over the fibre, which is what $\{F, G\}$ is above. ■